(i) Slope and intercept form: $(B \neq 0)$: The given equation can be written as

$$y = -\frac{A}{B}x - \frac{C}{B} \quad \Rightarrow \text{ slope} = -\frac{A}{B} \text{ and } y\text{-intercept} = -\frac{C}{B}$$

(ii) Intercepts form: The given equation can be written as

$$\frac{x}{\left(-\frac{C}{A}\right)} + \frac{y}{\left(-\frac{C}{B}\right)} = 1 \ (A,\ B \text{ and } C \text{ are all non-zero})$$

Comparing with intercept form, we get

$$x \text{ -intercept } (a) = \frac{-C}{A} \text{ and } y \text{ -intercept } (b) = \frac{-C}{B}$$

(iii) Normal form: Here A and B are not equal to zero,

Comparing $Ax + By + C = 0$, with $x\cos\theta + y\sin\theta = p$

We get $\frac{-A}{\sqrt{A^2+B^2}}x + \frac{-B}{\sqrt{A^2+B^2}}y = \frac{|C|}{\sqrt{A^2+B^2}}$

Here $\cos\alpha = \frac{-A}{\sqrt{A^2+B^2}}$, $\sin\alpha = \frac{-B}{\sqrt{A^2+B^2}}$ and $p = \frac{|C|}{\sqrt{A^2+B^2}}$

Example 6.19 Express the equation $\sqrt{3}x - y + 4 = 0$ in the following equivalent form:

(i) Slope and Intercept form
(ii) Intercept form
(iii) Normal form

Solution:

(i) Slope and intercept form

It is given that

$$\begin{aligned} \sqrt{3}x - y + 4 &= 0 \\ \Rightarrow y &= \sqrt{3}x + 4 \end{aligned} \tag{6.15}$$

Comparing the above equation with the equation $y = mx + b$, we have

$$\text{Slope} = \sqrt{3} \text{ and } y\text{-intercept} = 4$$

(ii) Intercept form

$$\begin{aligned} \sqrt{3}x - y + 4 &= 0 \Rightarrow \sqrt{3}x - y = -4 \\ \frac{-\sqrt{3}}{4}x + \frac{y}{4} &= 1 \\ \text{That is } \frac{x}{\left(-\frac{4}{\sqrt{3}}\right)} + \frac{y}{4} &= 1 \end{aligned} \tag{6.16}$$

Comparing the above equation with the equation $\frac{x}{a} + \frac{y}{b} = 1$

$$\text{We get, } x\text{-intercept} = -\frac{4}{\sqrt{3}} \text{ and } y\text{-intercept} = 4$$